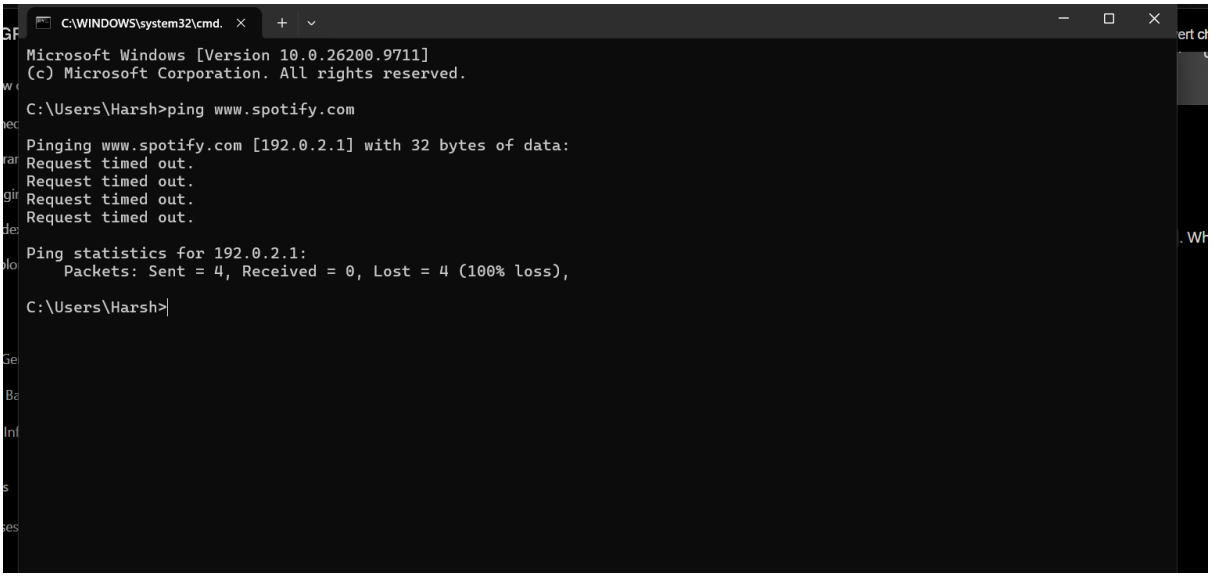
Day – 5

1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.



Steps:

1. Open Notepad as administrator and open C:\Windows\System32\drivers\etc\hosts.
2. Add the line 192.0.2.1 www.spotify.com and save the file.
3. Open Command Prompt and execute ping www.spotify.com.
4. Observe the error message and packet statistics.

Output: The command displayed Request timed out. four times, with 100% packet loss.

Result: The hosts file redirected www.spotify.com to the incorrect IP address 192.0.2.1, causing the ping requests to fail.

2 Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

The DHCP server assigned an incorrect default gateway (192.168.1.254) to the client device instead of the correct gateway (192.168.1.1).

Identification: The issue can be identified by checking the client's IP configuration using ipconfig /all and testing connectivity with the ping command.

Solution: Correct the default gateway in the DHCP configuration to 192.168.1.1 and renew the client's DHCP lease.

Conclusion: An incorrect default gateway prevents the client from communicating with external networks. Correcting the DHCP configuration restores proper network connectivity.

3. Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

DNS Troubleshooting and Solution

Troubleshooting Steps:

Open Command Prompt and run nslookup www.google.com to check whether the DNS server can resolve the domain name.

If the lookup fails, check whether the configured DNS server is correct and responding properly.

Use ping 8.8.8.8 to check internet connectivity.

Test DNS resolution using Google's public DNS server by running nslookup www.google.com 8.8.8.8.

If the public DNS server resolves the domain successfully, update the device's DNS settings to use a working DNS server.

Solution: Configure the network device or DHCP server to use a valid DNS server, such as Google's DNS (8.8.8.8) or Cloudflare DNS (1.1.1.1).

4. A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Troubleshooting an APIPA Address (169.x.x.x)

Possible Causes:

The DHCP server is switched off or unavailable.

The network cable is disconnected or the switch port is faulty.

The DHCP scope is exhausted, and no IP addresses are available.

The DHCP service is incorrectly configured.

A VLAN or network configuration issue is preventing communication with the DHCP server.

Steps to Resolve the Issue:

Check Network Connectivity: Ensure the device is connected properly to the network and the switch port is active.

Check the DHCP Server: Verify that the DHCP server is running and reachable.

Check the DHCP Scope: Confirm that free IP addresses are available in the DHCP pool.

Verify DHCP Configuration: Check the IP address range, subnet mask, and DHCP settings.

Renew the IP Address: Open Command Prompt and run ipconfig /release, followed by ipconfig /renew.

Check VLAN and Relay Settings: If the client and DHCP server are on different networks, verify the VLAN configuration and DHCP relay settings.

Verify the Result: Run ipconfig /all and confirm that the device receives a valid IP address from the DHCP server.

Conclusion: An APIPA address is automatically assigned when a device cannot obtain an IP address from a DHCP server. Checking network connectivity, server availability, DHCP scope, and configuration helps identify and resolve the problem.

5. Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Type	Issue	Symptoms	Possible Causes	Recommended Solutions
DNS	DNS resolution failure	Websites cannot be accessed by domain name but work by IP address.	Incorrect DNS server or DNS server unavailable.	Configure a valid DNS server, such as 8.8.8.8, and test using nslookup.
DNS	Incorrect DNS records	A domain name resolves to the wrong IP address.	Incorrect DNS record or hosts file entry.	Correct the DNS record or remove the incorrect hosts file entry.
DNS	Slow DNS response	Websites take a long time to load.	Slow DNS server, network delays, or DNS timeouts.	Check network connectivity and switch to a reliable DNS server if required.
DHCP	APIPA address assigned	The device receives an IP address starting with 169.254.x.x.	DHCP server unavailable or network connection failure.	Check cables, switch ports, DHCP server status, and renew the IP address.

DHCP	DHCP scope exhausted	New devices cannot obtain an IP address.	All available addresses in the DHCP pool are in use.	Expand the DHCP address pool or release unused leases.
DHCP	Incorrect gateway assigned	The device connects to the local network but cannot access external networks.	Incorrect default gateway configured in the DHCP scope.	Correct the DHCP default gateway and renew the client's IP lease.